DATA STRUCTURE
LAB PROGRAM

11. Write a C program to implement Stack operations such as PUSH, POP and PEEK

```
#include <stdio.h>
#define MAX 100

int stack[MAX];
int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Stack Overflow\n");
    } else {
        stack[++top] = value;
        printf("%d pushed into stack\n", value);
    }
}

void pop() {
    if (top == -1) {
        printf("Stack Underflow\n");
    } else {
        printf("Popped element: %d\n", stack[top--]);
    }
}
```

```
void peek() {
    if (top == -1) {
        printf("Stack is empty\n");
    } else {
        printf("Top element: %d\n", stack[top]);
    }
}

void display() {
    int i;
    if (top == -1) {
        printf("Stack is empty\n");
    } else {
        printf("Stack elements: ");
        for (i = top; i >= 0; i--) {
            printf("%d ", stack[i]);
        }
        printf("\n");
    }
}

int main() {
    int choice, value;

    do {
        printf("\n--- Stack Operations ---\n");
        printf("1. PUSH\n");
```

```
printf("2. POP\n");
printf("3. PEEK\n");
printf("4. DISPLAY\n");
printf("5. EXIT\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        printf("Enter value to push: ");
        scanf("%d", &value);
        push(value);
        break;

    case 2:
        pop();
        break;

    case 3:
        peek();
        break;

    case 4:
        display();
        break;

    case 5:
        printf("Exiting...\n");
```

```
                break;

            default:
                printf("Invalid choice!\n");
        }
    } while (choice != 5);

    return 0;
}
```

Output

```
--- Stack Operations ---
1. PUSH
2. POP
3. PEEK
4. DISPLAY
5. EXIT
Enter your choice: 1
Enter value to push:
22
22 pushed into stack

--- Stack Operations ---
1. PUSH
2. POP
3. PEEK
4. DISPLAY
5. EXIT
Enter your choice:
```